



Introduction to CS541 Artificial Intelligence

CS 541 Lecture 1

Zining Zhu



What is AI and why study it?

Definition · Applications · Impact

What is AI?

Intelligence demonstrated by machines, in contrast to the natural intelligence of humans and animals.

Systems that **perceive**, **reason**, **learn** from experience, and **act** toward a goal

Four classic framings: thinking vs. acting, humanly vs. rationally

- We take the **rational agent** view: act as well as you can, given what you know

Two toolkits carry the course

- **Learning from data**: supervised, unsupervised, language and multi-modal models
- **Search over structure**: BFS/DFS, Dijkstra, A*, game trees, MCTS

A moving target: symbolic AI → statistical learning → deep learning → foundation models

Once a problem is deterministically solved, it tends to stop being called AI, and gets called “automation” instead.

A brief history of AI in milestones

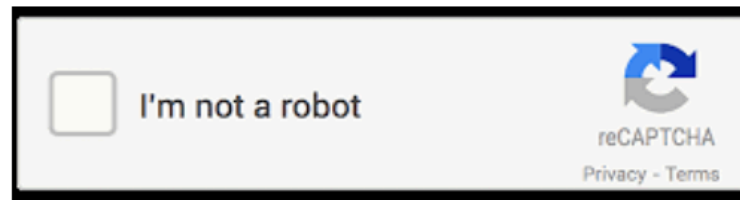
1950: Turing test



Alan Turing

1950: Can a computer convince a human that it is not a computer but a real person.

Now: Can a human convince a computer that he is a real person, not a computer



ProgrammerHumor.io

A brief history of AI (continued)

1956: Dartmouth conference and the term of “artificial intelligence” is coined.



A brief history of AI (continued)

1966: ELIZA, the world's first chatbot, was built.



A brief history of AI (continued)

1986: Backpropagation became popular

Letter | Published: 09 October 1986

Learning representations by back-propagating errors

[David E. Rumelhart](#), [Geoffrey E. Hinton](#) & [Ronald J. Williams](#)

[Nature](#) 323, 533–536 (1986) | [Cite this article](#)

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Abstract

We describe a new learning procedure, back-propagation, for networks of neurone-like units. The procedure repeatedly adjusts the weights of the connections in the network so as to minimize a measure of the difference between the actual output vector of the net and the desired output vector. As a result of the weight adjustments, internal 'hidden' units which are not part of the input or output come to represent important features of the task domain, and the regularities in the task are captured by the interactions of these units. The ability to create useful new features distinguishes back-propagation from earlier, simpler methods such as the perceptron-convergence procedure¹.

A brief history of AI (continued)

2000s: Deep Blue beats Kasparov, Watson wins Jeopardy.



A brief history of AI (continued)

2010s: Deep Learning beats human in ImageNet, AlphaGo beats Lee Sedol in Go.



A brief history of AI (continued)

2020s:

ChatGPT appears.

Claude Code appears.

AI affects entry-level jobs.

Major science problems claimed to be solved by AIs.

Many of us are using AIs.

AI in the world today

Language: assistants, translation, summarization, code generation

Perception: speech recognition, medical imaging, driver assistance

Decisions: recommendation, routing, fraud detection, ad auctions

Science: protein structure, weather forecasting, drug discovery

Games: chess, Go, poker, where search met learning

Nearly all of these mix **learning** and **search**, the two threads of this course.

AI in works

Automate repetitive tasks: apply for jobs, summarize documents

AI-assisted coding: “vibe-coding”, autocomplete, and code review

Polish texts: polish (and send) emails, essays, and reports



AI in social media

Recommendation system: Tiktok, YouTube, Instagram, RedNote, Twitter/X, LinkedIn, ...

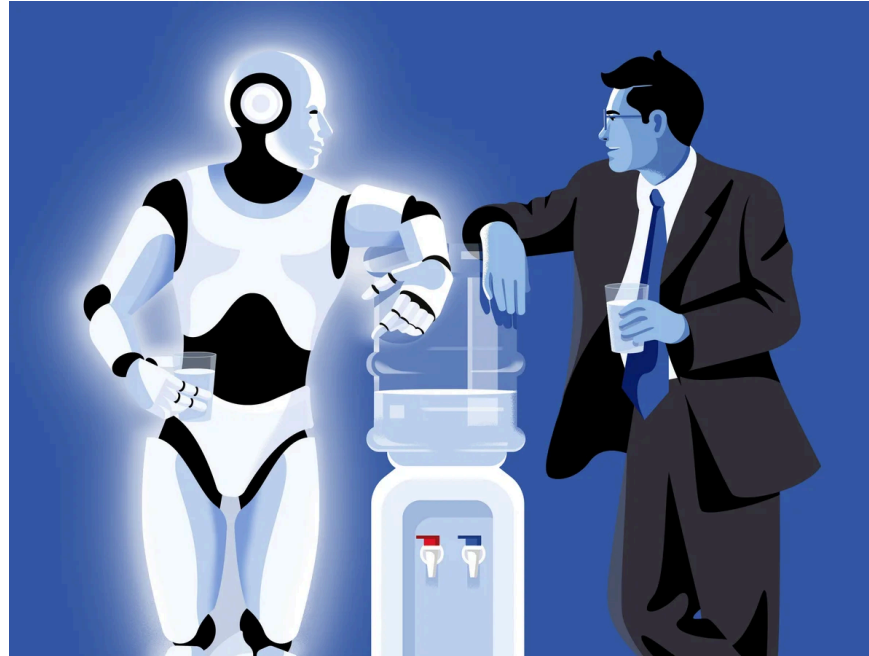
Content generation: text, image, audio, video, ...



AI in decision making

Fraud detection: detect fraudulent transactions, detect spam emails, detect fake news

Route navigation: search for shortest path, re-route when traffic is heavy

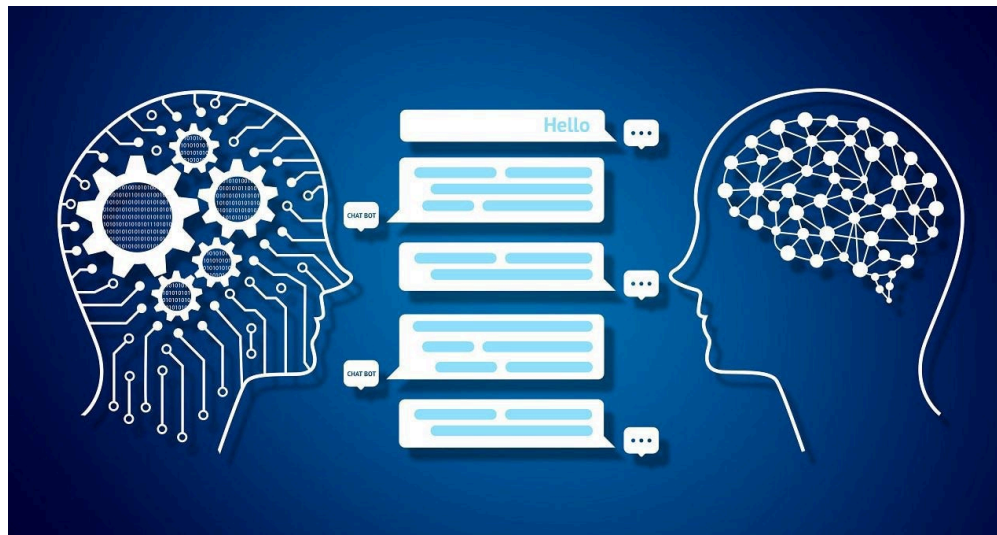


AI chatbots in daily lives

Website Q&A chatbots: customer service, technical support, and product recommendations

Daily smart assistants: Siri, Alexa, Google Assistant, and ChatGPT

Translate languages: Google Translate, DeepL, and Microsoft Translator



AI in education

Personalized learning: as “AI tutors”

Explanation: and multi-round explanations until we understand

Incentives: AI-generated motivational messages and learning plans

(Many also realize that AI can be a distraction and discourage thinking, so it is important to use it wisely.)

AI in scientific discovery

Predict protein structures and discover drugs: AlphaFold, AlphaDrug

Astronomy: detect exoplanets, classify galaxies, analyze telescope data

Physics: simulate physical systems, predict particle interactions, optimize experiments

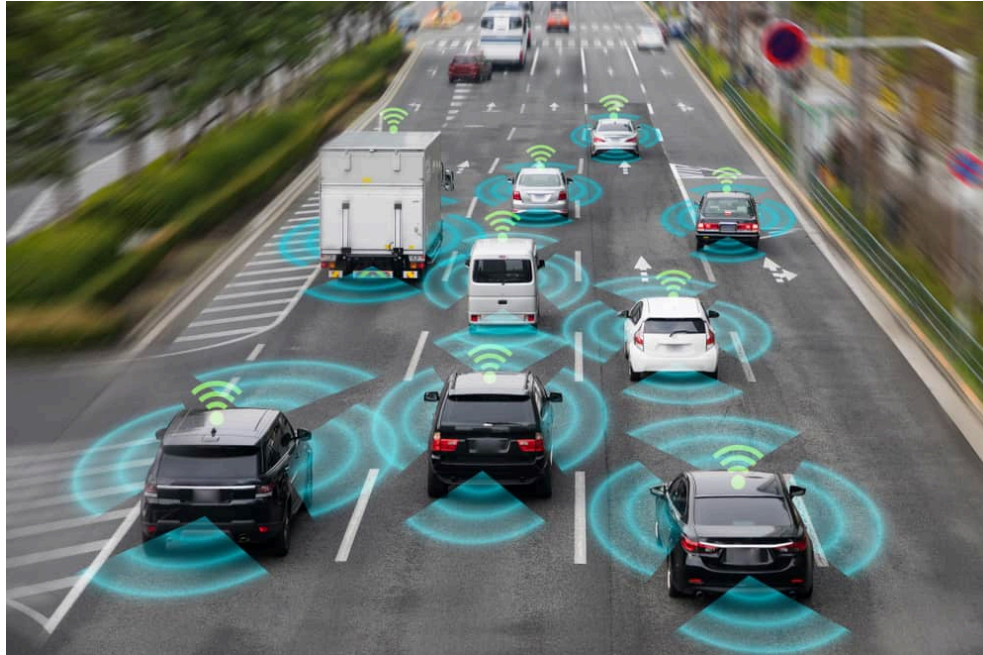
Climate science: predict climate change, model weather patterns, optimize energy usage

General scientific research: review literature, summarize papers, generate hypotheses, design experiments

AI in driving

Self-driving vehicles: Perceive the environment, make decisions, and control the vehicle

Driver assistance: Detect driver fatigue, lane departure, and collision avoidance



Why study AI?

AI has critical improvement areas:

Fairness: models inherit the patterns in their training data

Privacy: where data came from, what models memorize, who consented

Reliability: confident wrong answers, fabrications, synthetic media

Cost: large models carry a real compute and carbon footprint

Accountability: who answers for a bad decision? (EU AI Act, NIST AI RMF)

We should understand the tools **and their limitations**, so we can use them wisely and improve them.

Objectives of CS 541

After successful completion of this course, you will be able to:

1. Understand different AI models
2. Understand computational cost of AI algorithms
3. Being able to formulate a problem into an AI model (for example, machine learning, search)
4. Find & Apply AI algorithms to solve real world problems
5. Implement learning algorithms
6. Implement classic AI algorithms such as A* search



Course logistics

Schedule • Grading • Policies

Time and Location

Time: Thursdays 3:30-6:00 pm

Location: Babbio 203 and <https://stevens.zoom.us/j/91601429128>

Office hours: Wednesdays 10-12am, at GS429 or
<https://stevens.zoom.us/j/91601429128>

Prerequisites

linear algebra

probability

Python

About the Instructor

Zining Zhu, zining.zhu@stevens.edu

PhD 2024 from University of Toronto

Assistant Professor at Stevens Institute of Technology (2024-present)

Research area: explainable and controllable AI. A few interesting projects:

- Scientific claim-evidence reasoning
- Making AIs efficient and safe from fraud attacks
- Multi AI agent orchestration
- Computer-using AI agents
- HomeBotOne KOKO: Home robots

Website: <https://ecailab.org>

Teaching assistant

Ziruo Zhao, zzhao83@stevens.edu

PhD student at Stevens Institute of Technology

Research areas:

- human-AI co-author detection
- multi-agent orchestration

Tentative lecture plan, September to October

Date	Topic	Due
Sep 3	Overview of AI: definition, applications, history, impacts	
Sep 10	Supervised learning: bias-variance, logistic regression, optimizers	
Sep 17	Unsupervised learning: PCA, clustering	A1
Sep 24	Search: BFS, DFS, IDS, Dijkstra, A*, complexity	
Oct 1	Games: MCTS, SAT solvers, AlphaGo, AlphaFold	A2
Oct 8	Language models and inference	
Oct 15	Post-training: RLHF (PPO, DPO), RLVR (GRPO)	A3
Oct 22	LLM deployment: parallelism, ZeRO, RAG, agents	Teams

Tentative lecture plan, November to December

Date	Topic	Due
Oct 29	Reinforcement learning 1: problem setting, REINFORCE	A4
Nov 5	Reinforcement learning 2: actor-critic	
Nov 12	Project proposal presentations	Proposal
Nov 19	Multi-modal AI models: VLM, VLA	
Nov 26	Thanksgiving break, no class	
Dec 3	Project final presentations	Slides
Dec 10	Learning theory: risk, generalization bounds	
Dec 17		Report

Tentative assignment plan

Assignment 1: Supervised learning in Python

Assignment 2: Unsupervised learning

Assignment 3: Search

Assignment 4: Games

Scores breakdown

Component	Weight	Details
Participation	20%	Quizzes through Socrates platform
Assignments	40%	Four at 10% each, due 3:00 pm: Sep 17, Oct 1, Oct 15, Oct 29
Course project	40%	Proposal 10% · presentation 10% · report 20%

Teams form around Oct 22. Start thinking early about a problem you would actually like to solve.

Late policy

10% off per 24 hours late, down to a minimum score of 0%

Example: 85%, handed in 30 hours late → two blocks of 10% → 65%

After 10 days the deduction reaches 100%. Submit whatever you have before then; partial work still earns partial credit

Applies to assignments and project milestones alike

Talk to me **in advance** about emergencies, religious observance, or accommodations through the Office of Disability Services

Policy for Generative AI

You may use tools such as Gemini, Codex, or OpenCode to generate ideas and brainstorm

In course projects, you may use AI tools to generate code, but you must understand and be able to explain the code you submit

You may not submit work generated by an AI program as your own

If you include AI-generated material, **cite it**: any AI-generated work must be declared, with the level of use documented and the tool named (e.g., “Claude Code is used to proofread the codes and draft the analysis”)

Leaning on these tools also stifles the independent thinking this course is here to build
Plagiarism and cheating are handled under the relevant Stevens policies



Course project

Project overview

Two avenues:

1. Research project.
2. Application project.

Research project

Pick a research problem in AI. Some examples:

- How well can an agent playing an Atari game understand the game rules?
- The model XYZ does not correctly answer “Is 3.9 larger than 3.11”?
- A chatbot forgets my system prompt instructing it to speak in a given persona.
- When you say “improve it” to a model, it does not make the answer to a math problem better.

Research project (continued)

Solve it using new AI technologies.

Work on the problem, and either replicate an existing solution or propose a new solution. Evaluate, so that you can show your solution is better than existing solutions.

Starting point: read a recent paper from e.g., NeurIPS, ICML, ICLR, AAAI, ACL, CVPR.

- In the abstract, the sentence starting from “However” often points to the problem.
- The problem identified in a paper is usually too large for a course project; pick a subproblem.

Application project

Pick a real-world problem. Some examples:

- Meetings are noisy and complex, and you want to summarize them into your notes.
- You are planning a trip to a new city, and you want to optimize your itinerary.
- You want to keep up with the latest news on a topic, but you don't have time to read all the articles.

State the stakeholders (e.g., Who cares about the problem? If the solution leads to a product, who will be the users? Who will be impacted?)

Solve it using solutions involving AI.

Use both quantitative and qualitative metrics to evaluate your solution.

Starting point: read the news, or reach out to people who might be potential stakeholders.

Structure of the project

Form a team!

Work in a team of 2-3 students.

Teams of 4 students are allowed, but I'll expect the scope to be significantly larger.

Cross-section (CS-541 and CS-541-WS) teams are allowed.

Project steps!

The project takes four steps, each of around 2 weeks.

1. Specify the problem
2. Review existing solutions
3. Propose your solution
4. Evaluate your solution (especially: show it is better than existing solutions)

The above applies to both types of course projects.

Deliverables of the project

Proposal: 10', a short (1-2 pages) write-up.

- State the problem clearly.
- Review the existing solutions.
- Propose a solution.

Final presentation: 10', presentation.

- Review the problem.
- Describe your solution.
- Evaluate your solution.

Project report: 20', a report (≤ 8 pages + unlimited pages for references and appendix).

- Follow the project report template! Template will be provided in Canvas soon.



THANK YOU

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